

CAPTURE THE FLAG

THE CLIENT HANDBOOK FOR DR RORY HOPCRAFT

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INTRODUCTION



Welcome to your website – 'Capture The Flag'!



Out with the boring and in with the fun! This is a multiplayer game mixing recreation and learning. We aim to engage students through the art of video games whilst reinforcing their current education. This is done through a balance of rapid-fire questions mixed with a set of custom-made mini games.

Whilst the scoring system is dependent on their success of the mini games, students are encouraged to build on their knowledge base of a subject area to accumulate more points through a score multiplier. Each correct question gives you a higher multiplier value. More multiplier – more points!

We have aimed to motivate students as much as possible. And what better way to do this than a bit of friendly competition! Students are divided into two teams: A and B. The team with the highest score wins, so

friendly competition and collaboration are highly encouraged – sharing tips and pieces of information between each other.

COMMUNICATION/SUPPORT

Whilst the scope of this project ends on the 6th of May 2026, the team is willing to provide support, bug fixing and important features where feasible.

Contact information:

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Proceeding the mentioned project deadline, each member of the group will be on summer break from university and will be returning home. Whilst we will make every effort to respond to our student emails, please allow for a longer response time. If any development and/or bug fixes are needed, the time scope of this will be determined by the recipient's availability. If needed, work requested may be sent to another group member or amongst multiple developers to work on. Note that upon our return to our course in September, we will be available to answer questions more available, but any developments will be worked around our schedules.

WEBSITE SERVICES

This project contains four key, interconnected sections. These consists of the following:

SIGN-IN

- Every user of the website must do this to access other features of the website
- This is operated through google (as such, each user must have a google account to access the website)
- The user is then added to the websites storage area (firebase). Stored information is kept limited to keep the user's data private and secure. Example of users stored information:

Field:	Data:
Account created on	27 April 2026 at 10:15:05 UTC+1
Email	John.doe@gmail.com
Last login on	29 April 2026 at 16:20:02 UTC+1
Role	Student
User ID	Aa6uY2t0UTNWiQxotXSu4lbBV8u1
Username	John Doe

Table 1 – Firebase user example

LOBBY JOINING

- Players must input a lobby join code to play
- Teams A or B are selected along with a role (roles currently have no unique features so any are acceptable)
- A waiting screen is used in preparation of the game starting

THE GAME

- Sets of questions are displayed. Each question has four answers – three incorrect and one correct.
- Upon selecting an answer, it is either marked as the right or wrong selection. Upon successful answers, the player increases their score multiplier
- Mini games are also displayed here for the users to play. Score is built up through efficient gameplay and then increased by the multiplier at the end

LOBBY CREATION

- Accessible to the teachers only
- Can create new lobbies along with a unique join code
- Question pack uploading is available here for the students to run through

QUESTION GENERATION

- Accessible to the teachers only
- You can input any topic of your choosing. For example, physics.
- A custom set of questions are generated around this topic
- Questions are displayed in an editor, available to any alterations

GETTING SET-UP

A guide on setting up and accessing the website. Outlined here are the two methods: Local and on-campus.

LOCAL USE

To set up a local server, you must first set up your host device. You are required to have **two** servers running, one for backend connection to the AI model and one for access to other people on your network. To get everything working correctly, you need to have a few items installed. These are:

1. The website files themselves – [guide](#)
2. Nodejs – [guide](#)
3. Ollama – [guide](#)
4. Python(4.1) OR Visual studio(4.2) – [guide](#) OR [guide](#)

1. DOWNLOADING THE WEBSITE FILES

You have two options here. Either using a GitHub folder downloader or through a command prompt. When referencing the GitHub repo – you can find a link to this in the [appendices](#).

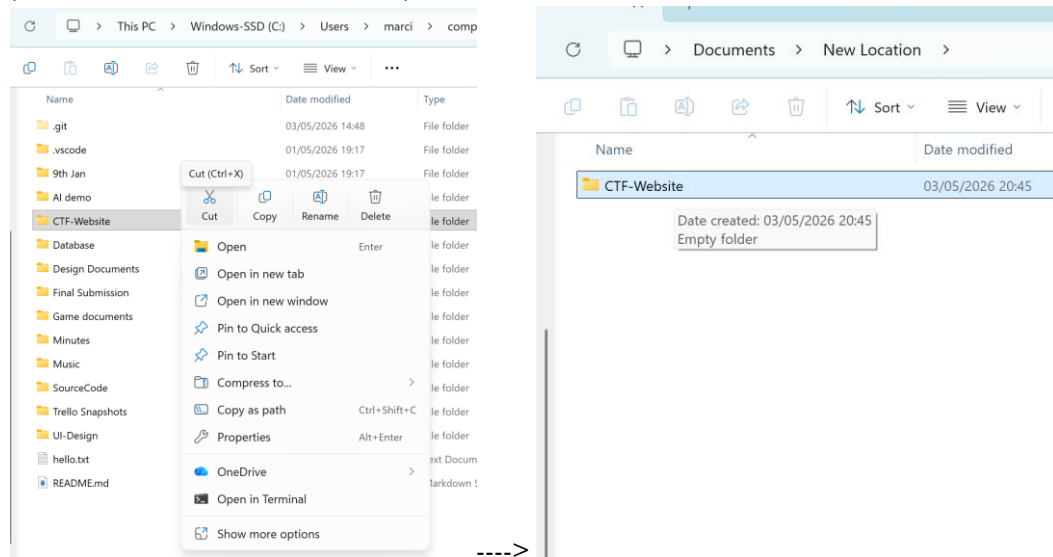
Option 1 – Using a GitHub downloader (the easier option):

1. Open a GitHub folder downloader. Can use any you wish but one I have used is: <https://download-directory.github.io/>
2. Insert **the folder link** (not the whole project repository, found in the folder **CTF-Website**). Download file. Open in file explorer
3. Extract the zipped folder into a location of your choosing

Option 2 – Using command prompt:

1. Open a terminal
2. Use the command: `git clone *repo link*`
(Be sure to use the whole repo link – not the CTF-Website folder)

3. Open downloaded file in your file explorer
4. Cut and paste the CTF-Website folder into a new location of your choosing
(You can now delete all other folders)



2. NODEJS

You also have two options here. Either downloading manually or through a terminal. **NOTE: Before you install node.js – check if you already have it installed** via the following:

1. Cd *project directory*
2. Node server.js

If you get an error displaying something similar to 'unable to find node' or 'node is not a recognised command' – proceed with the following download options.

Option 1 – Via their website:

1. Visit <https://nodejs.org/>
2. Click 'get node.js'
3. Follow download instructions

Option 2 – Via terminal:

Windows: 'winget install OpenJS.NodeJS' -> 'yes'

Mac: 'brew install node' (check [FAQs](#) on how to download Homebrew if brew is not a recognised command)

Linux: 'sudo apt update'
'sudo apt install nodejs npm'

3. OLLAMA

To download Ollama you again have two options. Through their website or via a terminal.

Option 1 – Via their website:

1. Visit their website here: <https://ollama.com/download>
2. Download the appropriate version for you – either Windows, Mac or Linux
3. Follow installation guide

Option 2 – Via terminal:

Windows: 'irm https://ollama.com/install.ps1 | iex'

Mac: 'curl -fsSL https://ollama.com/install.sh | sh'

Linux: 'curl -fsSL https://ollama.com/install.sh | sh'

4.1 PYTHON (IF NOT USING VS CODE)

This python server is executed through the terminal. You first need to check if python is installed. This is done through:

'python3 --version'

(Note this has a double '--' before 'version' and works on all platforms) (windows might use 'py --version').

If this doesn't display something similar to 'Python 3.11.6' then proceed with the following install steps:

Option 1 – Via their website:

1. Go to <https://www.python.org/downloads/>
2. Download version of python (current website build works on 3.14.4)
3. Follow install instructions

Option 2 – Via terminal:

Windows (PowerShell – NOT TERMINAL): 'winget install Python.Python.3'

Mac (Presuming Homebrew is installed - see [FAQs](#) if not): 'brew install python'

Linux: 'sudo apt update'
'sudo apt install python3 -y'

4.2 VS CODE (IF NOT USING PYTHON)

To install this, you need to:

1. Head to their website - <https://code.visualstudio.com/download>
2. Download appropriate version dependant on device type
3. Follow install instructions
4. Once installed:
 - a. Find the 'extensions' icon on the left and open it
 - b. Type in or find 'Live Server'
 - c. Install it
5. Click the cog icon in the bottom left corner and direct to: settings -> extensions -> live server config -> settings:host. Alternatively, open setting and type 'live server host' in the search bar.
6. Change this value to 0.0.0.0 as shown below

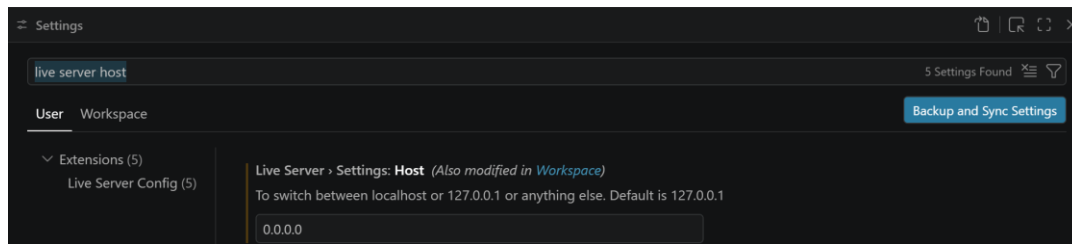


Image 1 – Host IP configuration

NOTE: This will make the website unavailable from localhost and must be accessed via IP (more on this soon)

ON-CAMPUS USE

Our group currently has a dedicated Virtual Machine on the university's servers. This is a Linux environment where we can host the website. This method is still currently under development as we have as issues accessing the website from outside the machine. This is likely due to accessing rules at are in place, but we are working on fixing this.

When this is fixed, you can access the website through *link will be provided when working*. This is only to be able to be accessed when operating under the university's networks. Alternatively, you can connect to their network via 'FortiClient' VPN:

<https://liveplymouthac.sharepoint.com/sites/WorkAtHome/SitePages/FortiClient.aspx?xsdata=MDV8MDJ8bWFyYy5qb25lc0BzdHVkZW50cy5wbHltb3V0aC5hYy51a3w3MWVhMDVhZmFkNGU0YTQwMDNiOTA4ZGU2ODgxNDNkY3w1NDM3ZTdYjgzZml0ZDFhYmZkM2JiMjQ3ZTA2MWJmMXwxZmFkMDYyZTAxNjU0NDc1MDI2fVua25vd258VfDgcGJHWNiM2Q4ZXIKRmJYQjBiVTFoY0draU9uUnlkV1VzSVWxZaU9pSXdmakF1TURBd01DSXNjBfEPt2IKWGFxNHpNaUlzS5WtGT0lqb2lUyOZwYkNjC0IsZFYJam95ZIE9PpxwffHx8&xsdata=RkJsRThGMGV0QTNhMFpwbCtfczdra1h3VH8aeXFXUDV5YjJ4cmJEMk9rQT0%3D&CT=1777909321178&OR=QWA-NTB-Mail&CID=f51ccb9-62b7-b463-3997-a593a70138a5&clickParams=eyJYLUFWcE5hbWUiOiJNaWNyb3NvZnQgT3V0bG9vayB8ZWlqQXBwliwiWC1BcHBWZXJzaW9uIjoiajAyNjA0MjQwMDUuMTQiLCJPUyI6IldpbmRvd3MgMTEifQ%3D%3D>

The end goal of getting this operational is to avoid having to go through the lengthy and tedious process of setting up for full use – having the website online for as long as needed.

GETTING THE LOCAL WEBSITE ONLINE

No matter which server method you are using (python or VS) you will always need to complete these steps:

1. Open a terminal
2. Find your current IP address via:
 - a. **Windows:** 'ipconfig' (Note the IPv4 Address under the Wireless LAN adapter Wi-Fi)
 - b. **Linux:** 'ip a' (Note the – inet 156.143.x.x – value)
 - c. **Mac:** 'ipconfig getifaddr en0' (Note the – 156.142.x.x – value)
3. 'cd *project directory*'
4. 'node server.js' (should receive a message saying 'Backend running at http://0.0.0.0:3000')

If using Python implementation:

5. Open a new terminal (**keep the old one running**)
6. 'cd *project directory*'
7. 'python 3 -m http.server 5500' (windows may need 'py -3 -m http.server 5500')

If using VS Code implementation:

5. Open the folder via: file -> open folder -> *Open project folder*
6. Click the 'go live' button in the bottom right hand corner. Shown below

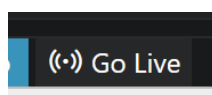


Image 2 – Go live button

Note: This will automatically launch the website and this site will display 'this site cant be reached'. You can ignore this and close that webpage down if you wish

ACCESSING THE LOCAL WEBSITE FROM OTHER DEVICES

Right, you should now have everything online and operational! On to actually using the website! To achieve this, you much make sure that all devices you would like to use this are all operating on the same IP address. See '[getting the local website online](#)' [step 2](#) on how to do this. **THE WEBSITE WILL NOT BE ACCESSABLE IF DEVICES ARE UNDER DIFFERENT IPS.** Once you have ensured this, simply type the following website address into the browser:

`http://*IP address*:*Port number*`

This will look like this:

`http://10.254.170.233:5500`

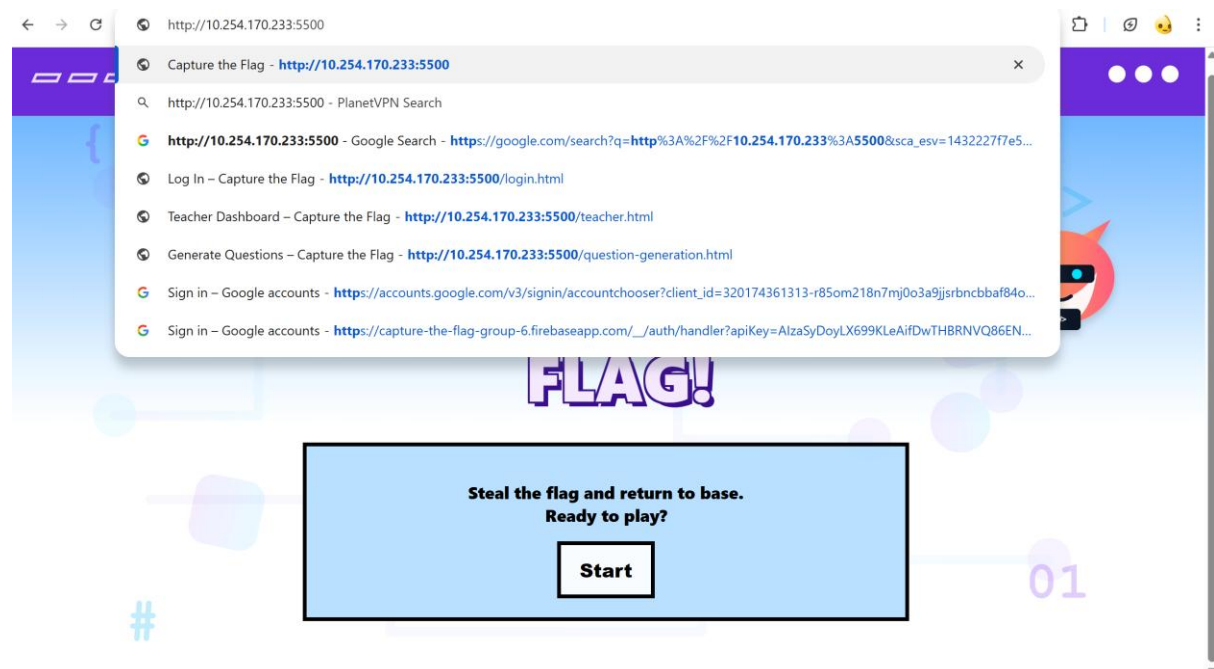


Image 3 – Example URL

USING THE WEBSITE

The website is set up to be used by two primary parties: students and teachers. As such, there are two separate sections to the website to operate accordingly. To access these – navigate to the 'index.html' file and open it. You will be presented with the introduction screen. Click the 'start' button to proceed.

You will then be prompted to log in using google. If you have not previously used a google account on this website, this will automatically create your profile from you google account details. Select the appropriate student or teacher buttons to proceed forward.

TEACHERS - LOBBY

Once logged in, you are able to create a lobby for your students to join. Press the 'Create new session' button to do so. This will bring you to the lobby management screen. Here, you can:

7. View the join codes to share with your students. This should be shared with each student who would like to play.
8. View the current status of the lobby. There are three states:
 - a. **Open:** For setting up the lobby. This is the default state when creating a new session. Students can join in this state
 - b. **In progress:** When the game is in session. This is set when clicking the 'start match' button.
 - c. **Closed:** When the game has concluded. The students can no longer join or play when in this state.
9. **Start or end the match.** Only use these once all students are ready. The start button will change the lobby state from 'open' to 'in progress'. Students that have selected their roll will be in a ready state – looking at a screen saying, 'waiting to start'. Starting the match will put them into the game. The end match button will change it to 'closed'.
10. **Create a new session.** For when you would like to end the current session and start a brand new one. Will take you back to the 'create new session' screen.
11. **Upload question files.** Where to upload your question files. These files must be uploaded as a JSON file. To create or edit these question packs – head to the 'generate new question files' button. These questions are stored to the current lobby for the students to answer.
12. **Generate new question files.** This will take you to the question generation and JSON editor page. More information will be shared below on this.
13. **View students currently in your lobby.** This will show each student along with their team in the current lobby.
14. **Live team scoring.** This is only available to view once the match is 'in progress'. Teams A and B are displayed side by side and scores will be updated in real time as the students accumulate score.

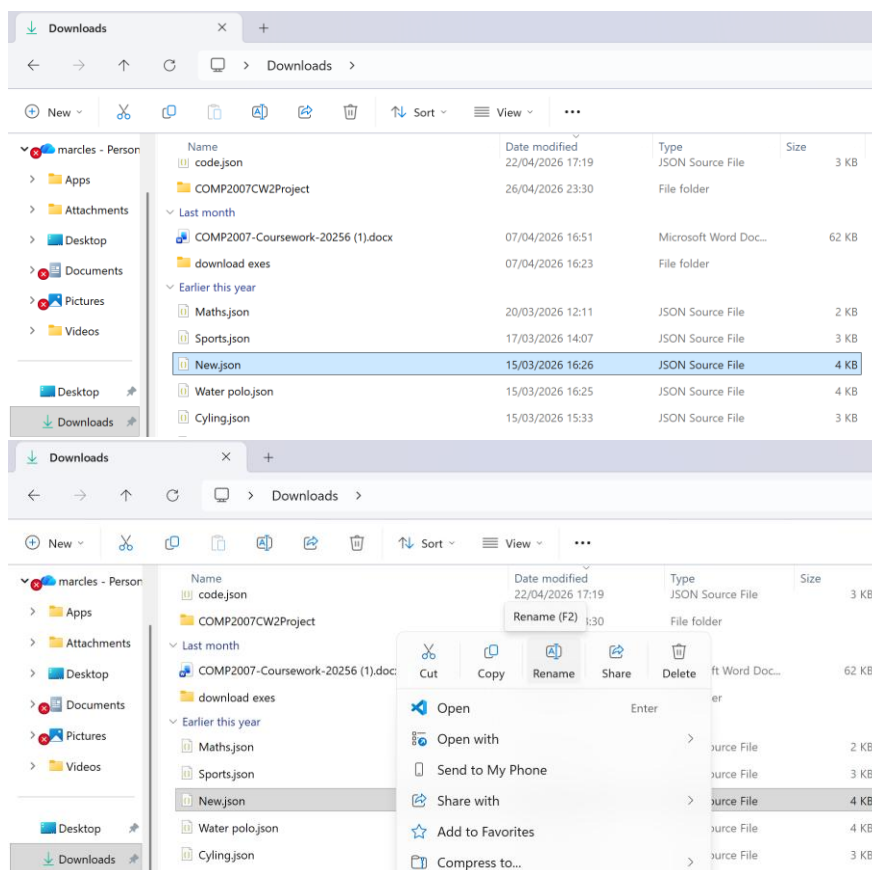
TEACHERS – QUESTION GENERATION

Upon clicking the 'Generate some!' button, you will be re-routed to the question generation page. It is here that you can create your own question packs for the students to use. All you have to do is input a topic. The AI bot will take this topic and create you a set of questions and answers. The questions will typically take around twenty to sixty seconds dependant on a few factors. These will automatically be displayed below when the AI has finished processing your topic. **IT IS IMPORTANT TO NOTE:** You will almost always need to change one to several of the correct answers here. Of the four answers that the bot will give you for each question, at least one of them should be correct but you may need to change this in the 'correct answer' field as it can get this wrong sometimes.

You are also able to add new questions yourself here from scratch. **IT IS ALSO IMPORTANT TO NOTE:** Questions used by the students are taken from the question packs in sets of four. As such, if you have nine questions total, only the first eight will be used. To access this ninth question, you must create questions ten, eleven and twelve.

To edit an existing question pack (if you would like to change some details or add new questions to it), you can select the 'choose file' button to upload the pack you would like to change. Here, this pack will be displayed and you are free to make any changes you like.

To save any changes, simply select the 'save' button displayed at the bottom of the page. This will create a new file that will automatically be saved to your downloads folder (in your computers file explorer). It is recommended to rename this file so you can differentiate between different question packs by right clicking the file -> rename.



Images 1 & 2 – File rename tutorial

STUDENTS – THE GAME

Once logged in as a student, you will follow a linear path to join and play the game. An explanation follows:

1. **Join the lobby.** You are required to input the join code provided by the teacher (covered in the 'Teachers – Lobby' section). This will allow all students to play in the same lobby.
2. **Select your team.** You can choose between teams A and B. To allow for a fair game, the number of students in each team should be equal. You are also required to pick a role but this currently has no effect on gameplay. Select 'Join Game!' to continue.
3. **Confirm team.** Select either: 'Yes, Join Game!' or 'Change Team/Role'. The first option will confirm you are happy with your selection and proceed you forwards. The second option will return you back to step 2.
4. **Wait for the game to start.** Whilst the lobby remains in an 'open' state, this screen will be displayed. The teacher should change it to 'in progress' to proceed.
5. **Answer questions.** Four questions will be displayed, answering each one correctly rewards you with points and a bonus mini game multiplier.
6. **Play mini game.** You will need to pay the mini game and accumulate the most number of points you can by successfully meeting that games objectives. Final score for this round is added to your teams total and the number of points awarded is the following: (points awarded from questions + (mini game score * multiplier)).
7. **Repeat 5-6** if four or more questions are left in the question pack.
8. **Game ends.** You are returned to 1.

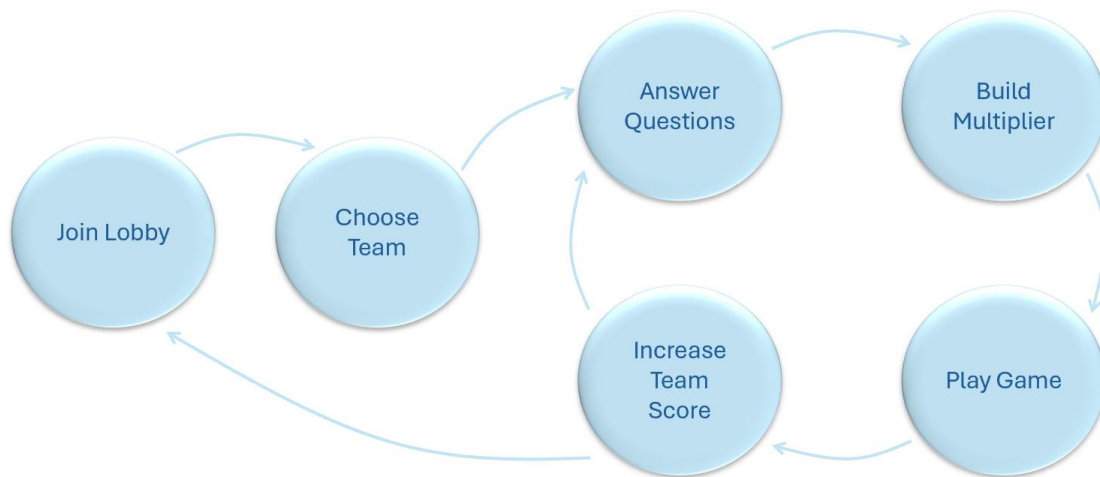


Image 3 – Game flow

FAQS

SIGN-IN

1. Error 'Sign-in failed. Please try again' showing immediately
This is likely due to the URL not being whitelisted in firebase. To fix, head to the firebase -> authentication -> settings -> authorised domains -> add domain. Copy and paste the domain currently being used into the box and click 'add'. Only needed if on a local server setup.

MAC USERS

1. Brew is not a recognised command. To download Homebrew follow these instructions in a terminal:
2. `"/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"`
3. `'echo 'eval "$(/opt/homebrew/bin/brew shellenv)"' >> ~/.zprofile'`
4. `'eval "$(/opt/homebrew/bin/brew shellenv)"'`
5. `'brew --version'` (to confirm installation)

APPENDICES

Whole GitHub link - <https://github.com/Plymouth-University/comp2003-2025-2026-group-6>

Project firebase: <https://console.firebase.google.com/u/1/project/capture-the-flag-group-6/overview>. Sign-in details to the google account are available upon request.

CLIENT SIGNOFF

Name:

Date:

Signature:

Declaration:

☐ By checking this box, I confirm that I have read and understood this handbook. I acknowledge that I am satisfied with the project as delivered at this stage and agree to adhere to the communication guidelines outlined in the [communication segment](#).